

Design of Experiments

How to decide what data to collect to answer your research question

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Created using Claude Code

Where you are in this workshop

- **Yesterday afternoon**

how to preserve data — data stewardship

- **This morning**

how to generate data on purpose — design of experiments

- **Shared premise**

your data is more valuable than you think; plan for it up front

Introduction to Statistical Thinking for Science

1. Statistical thinking

Signal, noise, factors, measurement. This is the language every DOE assumes you already speak. We'll build it up from EDA and the basic variance decomposition.

the foundation

2. The DOE family

OFAT, factorial, fractional factorial, response surface, active learning. Different tools for different objectives — and what happens when you use the wrong one.

the toolkit

3. Analyzing your results

Fit the model, check whether the effect is real, interpret interactions, verify. The steps that turn a table of numbers into a defensible answer.

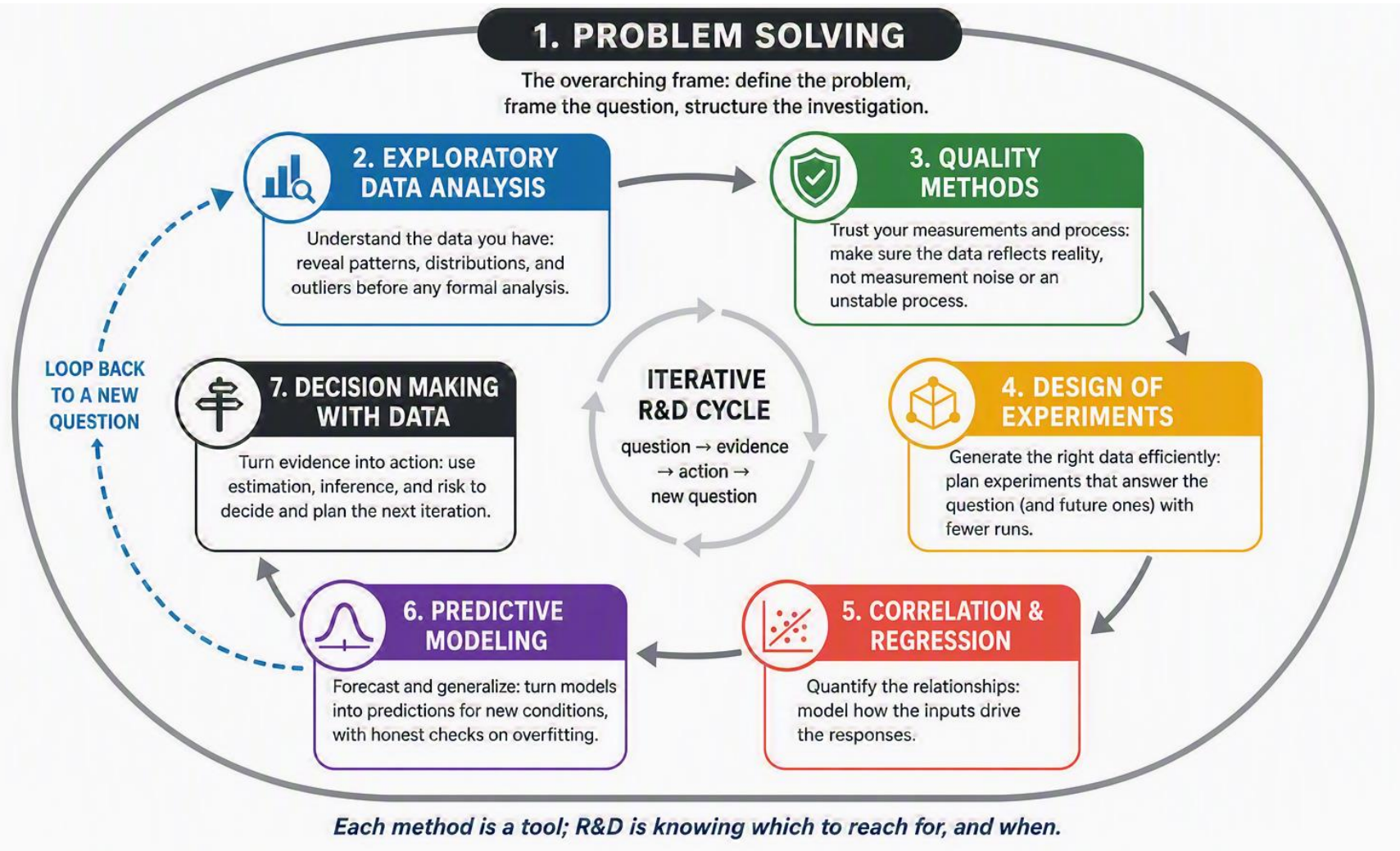
closing the loop

PART 1

Statistical thinking — the foundation

Signal, noise, measurement, and what an experiment is trying to accomplish

The R&D Problem Solving Toolkit



- Frame the question
- Generate trustworthy evidence
- Model what matters
- Decide what to do next

Foundation 1: Understanding data

HOW DATA IS BORN

Physical phenomenon

The underlying data-generating system that exists in the world. It responds to external conditions based on physics. This is usually what we're trying to study.

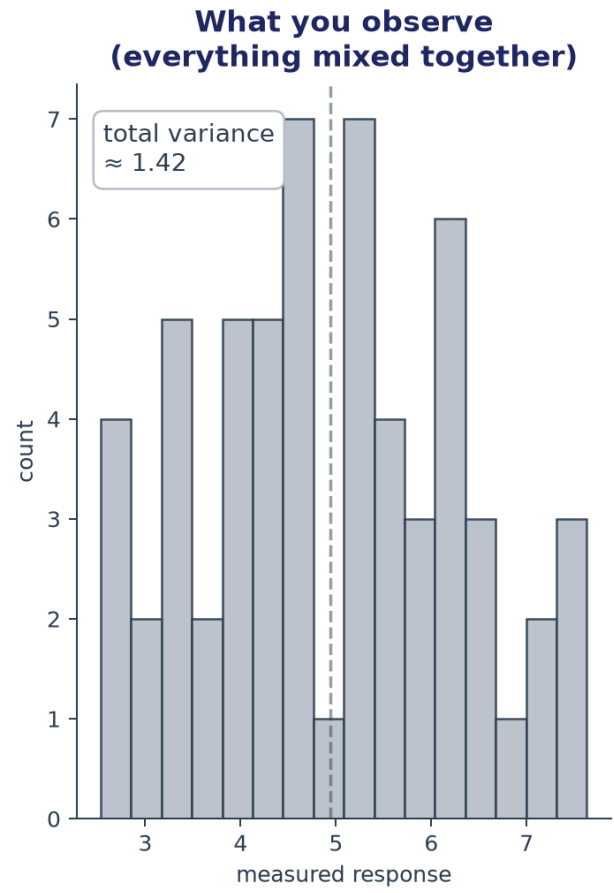
Structural noise

On top of the physical phenomenon, we have "other things that happened in the world that we aren't strictly interested in." Batches, operators, days, instruments, seasons.

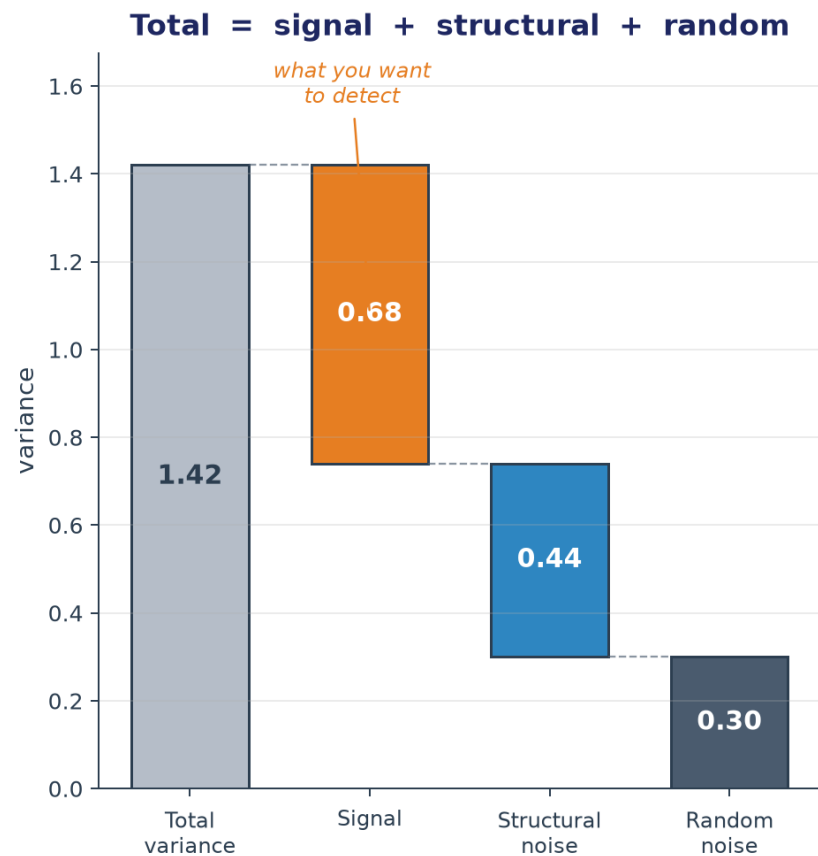
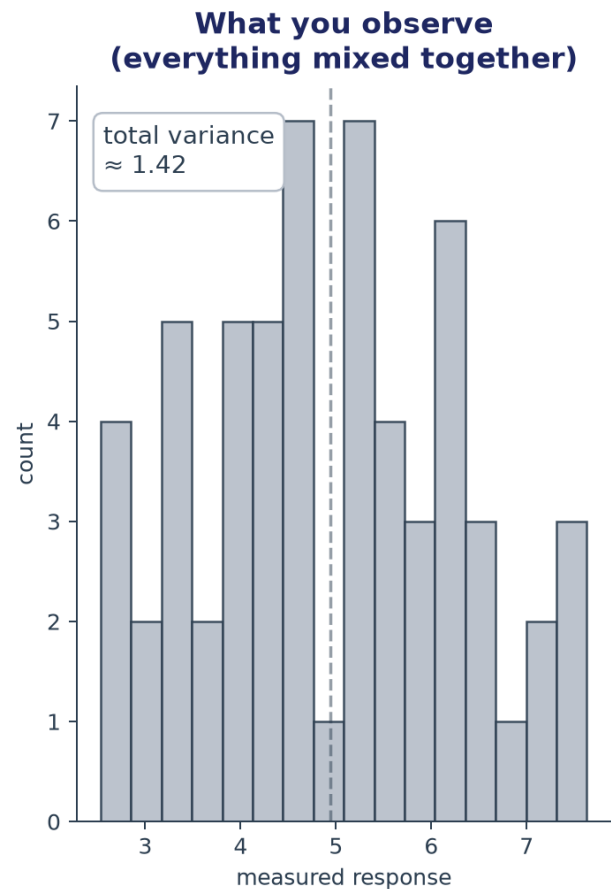
Random noise

Irreducible system, process, and measurement variability.

Data – measured observations – are a combination of all 3



Data – measured observations – are a combination of all 3

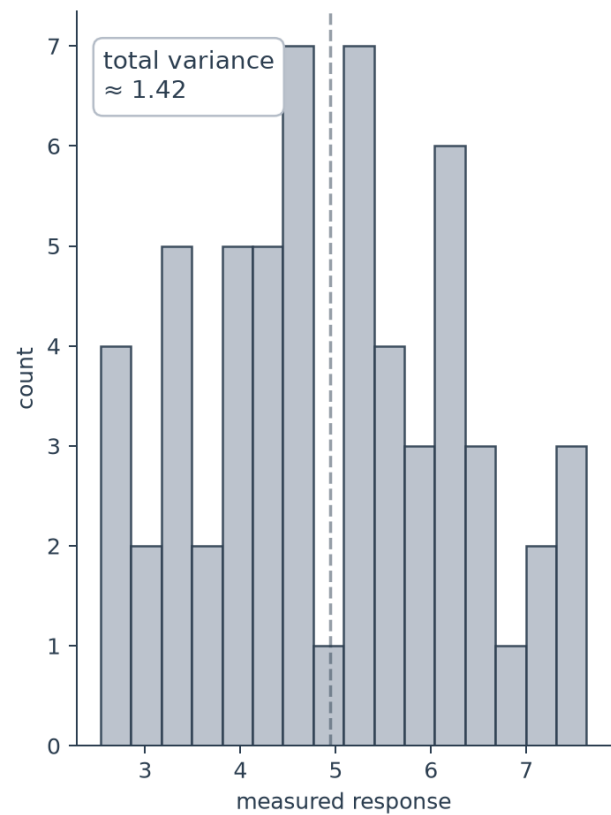


Total variance = signal + structural + random. Design is how we keep the signal from getting mixed up with the other two.

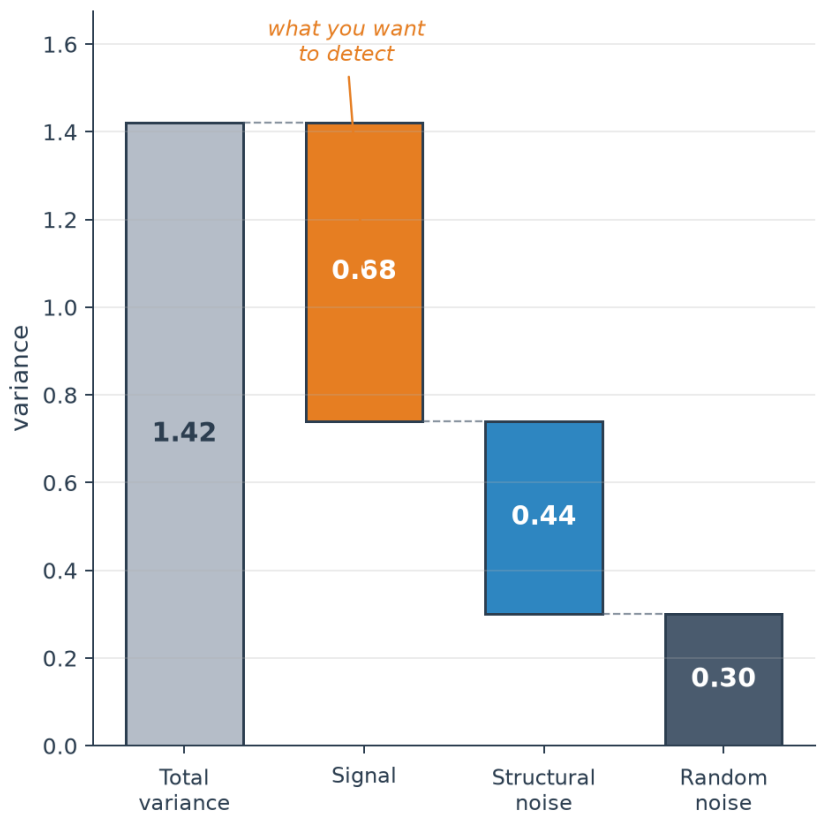
Data – measured observations – are a combination of all 3

Three sources of variation — and what design does about them

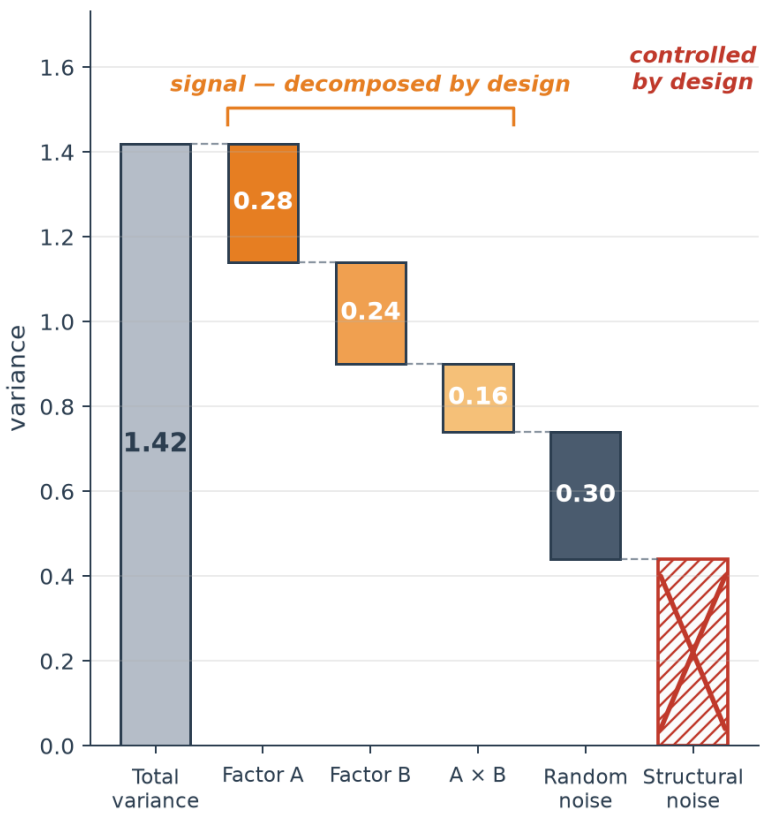
What you observe
(everything mixed together)



Total = signal + structural + random



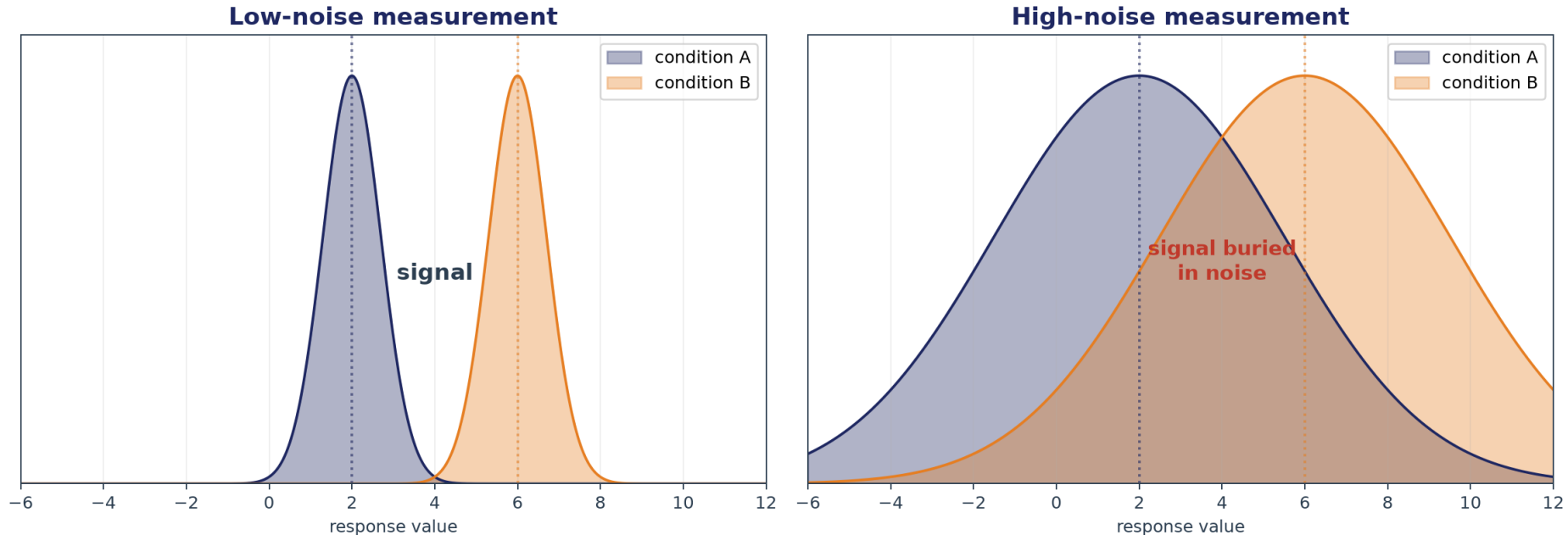
In a designed experiment:
signal decomposed, structural controlled



Total variance = signal + structural + random. Design is how we keep the signal from getting mixed up with the other two.

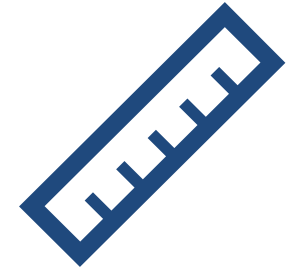
Foundation 2: Separating signal from noise is the goal

Same effect, two measurement systems



Same real effect, two experimental systems. On the left, the effect is clear and easy to see. On the right, the same effect is drowned in variability. This is the central problem statistical thinking is designed to address: how do we tell what our factors-of-interest did from what the day, the instrument, the operator, and the atmosphere did on top of it?

Measurement systems analysis – Know your ruler



TWO OPERATORS TITRATE THE SAME REACTION. WHICH ONE IS RIGHT?

Same reagent lot. Same protocol. Same reaction, sampled at the same time. Operator A calls the endpoint at 3.15 mL. Operator B calls it at 3.28 mL.

Neither is wrong. Both numbers are the reaction PLUS the operator's variability PLUS whatever else happened on that bench that day — a slight tilt in the burette, a difference in what two eyes see, a slightly different reference standard.

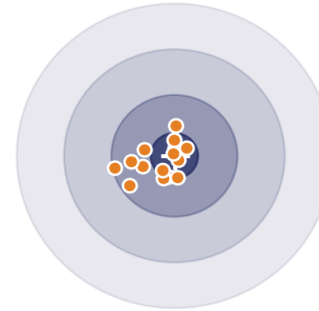
You cannot answer 'which is right?' until you've decomposed the variability into its sources. That is what a measurement systems analysis is for.

Precision vs accuracy

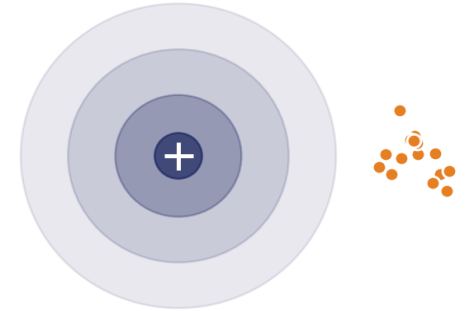
Precision is repeatability — the same person doing the same measurement gets a tight cluster around some value. Accuracy is whether that cluster is centered on the truth.

Imprecision can be addressed by replicates — the mean of 10 messy points is more accurate. You cannot replicate your way out of bias — biased means converge to the biased answer. Bias has to be detected and removed.

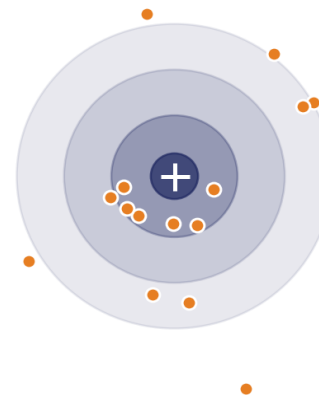
Accurate & precise



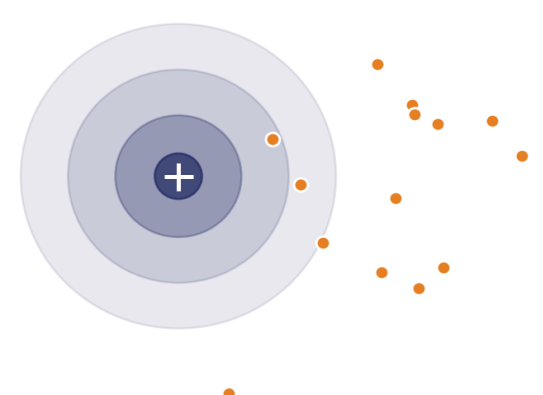
Precise but biased



Accurate but imprecise



Biased and imprecise



What MSA checks — four dimensions

Repeatability

Same operator, same sample, same day — do you get the same number? If not, the process is noisy at the operator level.

Reproducibility

Different operators, different days — is the number stable? This is where the two-operator titration story lives.

Bias

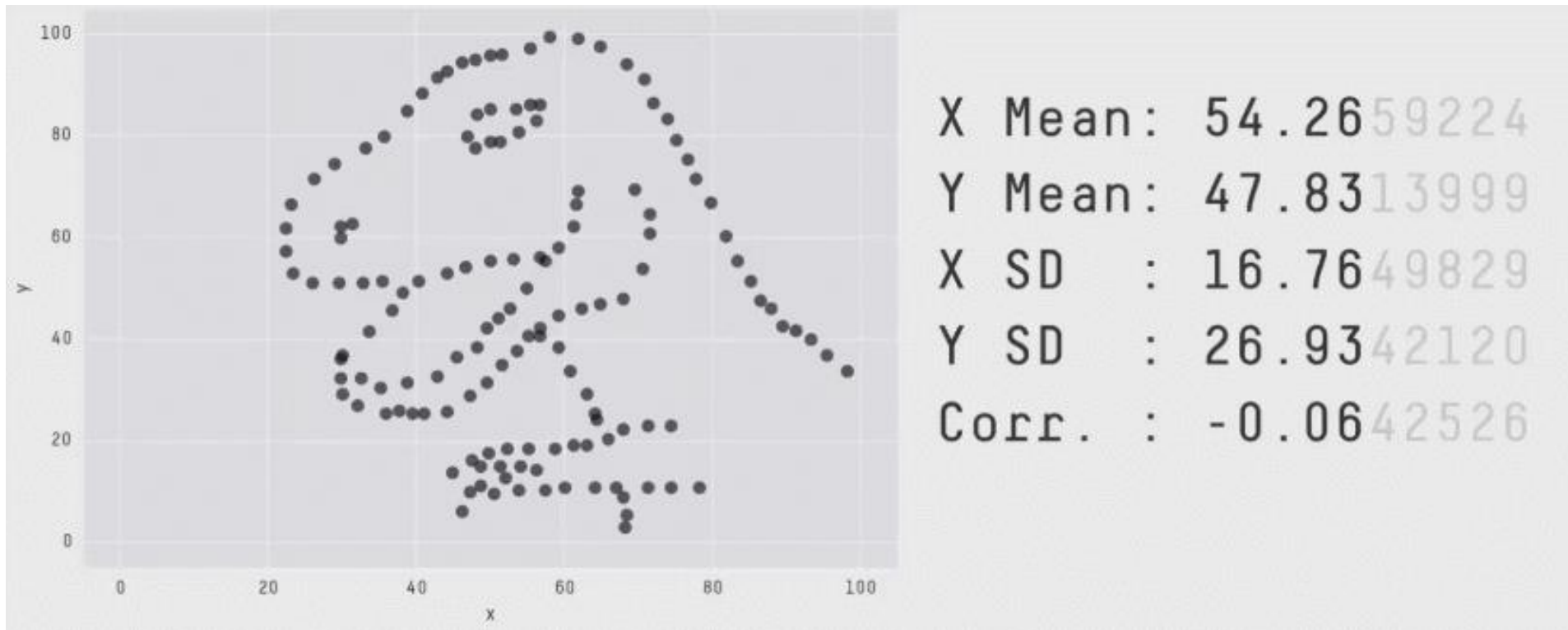
Is the average right, or shifted from the true value? Replication doesn't fix bias; a reference standard does.

Linearity

Does the measurement error stay the same across the range? Instruments that drift with magnitude create trends that look like real effects.

Foundation 3: Look at your data

The [datasaurus dozen](#) – 12 datasets with the *same* summary statistics



Exploratory Data Analysis (EDA) – a data scientist's lab

LOOK AT THE DATA WITH EACH OF THESE LENSES

Histogram

one variable at a time — is it symmetric, skewed, bimodal? Are there outliers?

Boxplot

compare distributions across groups — spread, median, and outliers side by side.

Scatterplot

two variables together — is there a relationship? Is it linear, curved, or absent?

Run chart

the measurement in the order it was collected — time-drift, sudden jumps, seasonality.

Correlation heatmap

many variables at once — quick screen for pairs that move together.

Residuals vs fitted

after any model, plot what's left over against what you predicted. Patterns here mean the model is missing something.

Statistical vs practical significance

Statistical significance asks: is the effect big enough that noise could not plausibly have produced it by chance? Practical significance asks: **is the effect big enough to matter** for the decision I'm making?

FOR EXAMPLE

A 0.2% yield improvement, measured across 500 runs, can be statistically significant but practically useless. A 5% improvement measured on 6 runs can matter enormously even if the p-value is 0.09. Report both cases, put your findings in their statistical context, and move forward with your decisions.

PART 2

The DOE toolkit

Where designs fit in the scientific method, and how to choose one

DEFINITION

Design of Experiments (DOE)

A structured approach to **planning experiments** so the resulting data can answer **specific questions** about a system — while using the fewest runs and separating real effects from noise.

FOR EXAMPLE

'What are the main effects of temperature, catalyst loading, and residence time on yield, and do any of them interact?' — with an experimental plan that will actually answer that.

The language of DOE

FACTORS (X)

- temperature
- catalyst loading
- solvent
- cure time

$$Y = f(X) + \epsilon$$

process / experiment

$\epsilon = \text{irreducible variability}$

RESPONSES (Y)

- yield
- selectivity
- impurities

Everything DOE does is designed to estimate f .

Factors (X), a.k.a. independent variables, are things you change on purpose. Responses (Y) are things you measure. The function f is the underlying reality you're trying to learn. Every measurement you take is that reality plus a bit of noise.

The empirical model most DOE aims to fit

$$Y = \beta_0 + \sum \beta_i x_i + \sum \beta_{ij} x_i x_j + \sum \beta_{ii} x_i^2 + \varepsilon$$

Y	response — the thing you measured (yield, viscosity, tensile)	x_i	factor i, coded to a common scale (usually [-1, +1])
β_0	intercept — the response at the design center	β_i	main effect of factor i — the slope in x_i direction
β_{ij}	two-way interaction between factors i and j	β_{ii}	curvature term — quadratic in factor i
ε	residual error — everything the model doesn't explain		

Your scientific hypothesis is encoded into this functional form. Then, you use statistics to test if $\beta \neq 0$

Your question → model choice → design

If you pick the wrong tool for the objective you have, running more experiments won't recover it.

What are you trying to learn?

Screening (which factors matter)? Optimization (where's the peak)? Mechanism (why)?

What model shape can answer that?

Linear main effects? Response-surface quadratic? Mechanistic differential equation?

What design supports that model?

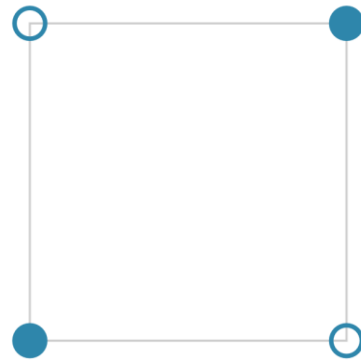
Fractional factorial screens. CCD or Optimal for surfaces. Targeted kinetic runs for mechanism.

Your design determines what you can learn

Match the design to the question: screen, characterize, optimize

As the investigation narrows, the factor set shrinks and the model grows.

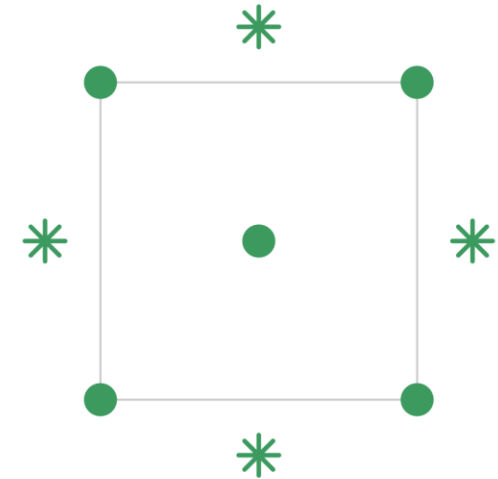
Filled/open = a fractional factorial's half-fraction; * = axial points that add curvature to a response-surface design.



factors ↓ · model ↑



factors ↓ · model ↑



1. Screen

Which factors matter at all?

Many factors → few

Model: main effects

2. Characterize

How do the key factors act & interact?

The vital few

Model: + interactions

3. Optimize

Where is the best setting?

Usually 2-3 factors

Model: + curvature

Your design determines what you can learn

To see these, a DOE will...

- + a main effect**
vary that factor across its range
- + an interaction**
vary two factors JOINTLY, not one at a time
- + curvature, i.e. a quadratic effect**
use ≥ 3 levels, or add center-point runs
- + signal vs noise**
replicate — rerun at the same conditions, not test the same sample twice
- + day/batch drift**
block on the drifting variable and randomize inside blocks

X DOE Mistakes

- interaction with fixed factor**
One factor at a time freezes other factors — interactions invisible
- curvature at 2 levels**
you fit a line through 2 points; curvature is undetectable
- no noise without replication**
one number per condition — can't separate signal from ϵ
- drift without randomization**
time trends read as fake main effects

Four ways people actually run experiments

Informed search

Chemistry intuition guides the next runs. Fast when intuition is right; has blind spots and is hard to audit.

One factor at a time (OFAT)

Vary one factor at a time, freezing everything else. What everyone does by default. Misses interactions completely.

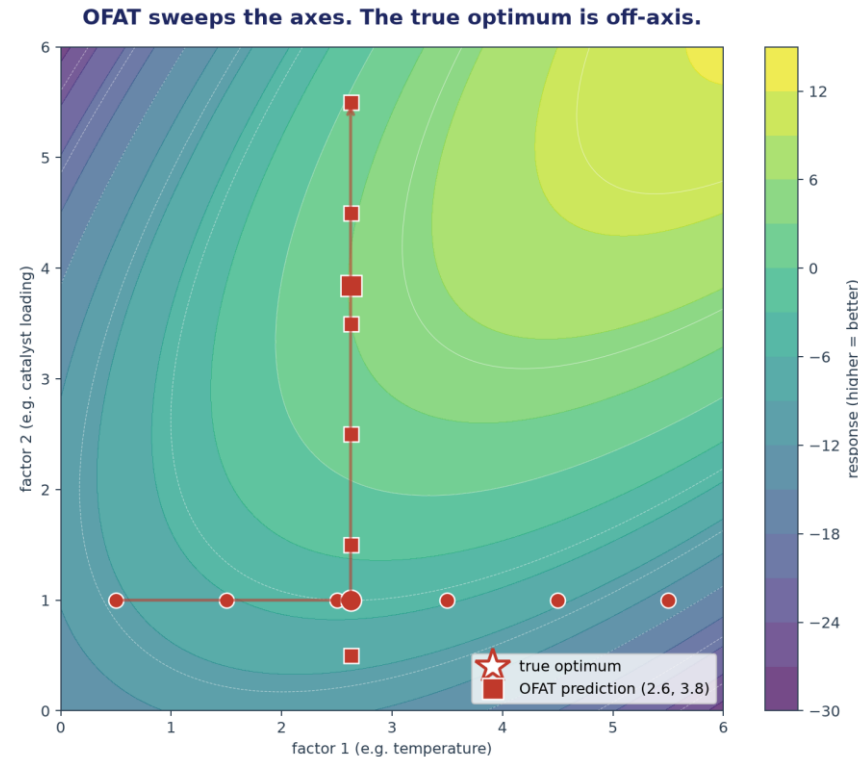
DOE

Plan the whole set of runs up front. Every main effect and every interaction, quantified — the design decides which β terms are estimable.

Active Learning

An iterative strategy to choose the next *most informative* experiment, i.e. will most reduce uncertainty or best improve target

A visual explanation of one-factor-at-a-time



OFAT freezes each factor at its per-factor best and then moves to the next. If the factors interact, e.g. if the best temperature depends on which catalyst you're using, that per-factor series never explores the corner where the interaction lives. OFAT looks reasonable when factors are truly independent. In chemistry, factors are rarely independent.

The DOE family (a survey, not a how-to)

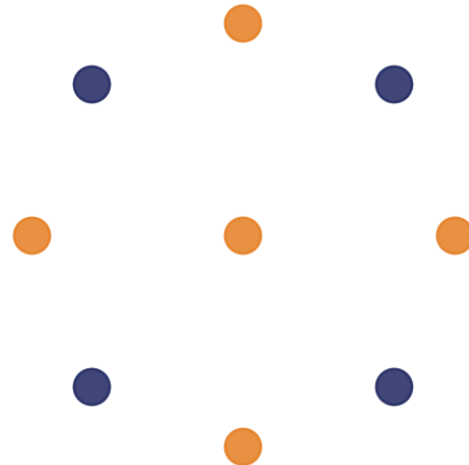
Every DOE family has its own shape — pick to match your objective, not the other way round.

Screening



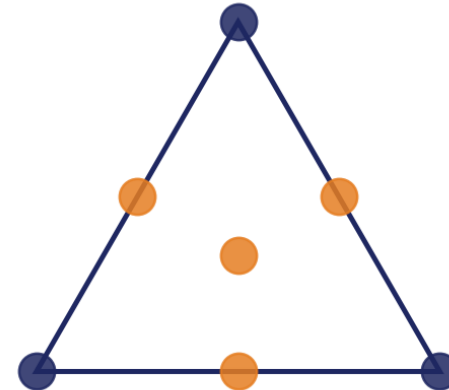
fractional factorial
Plackett-Burman

Response-surface



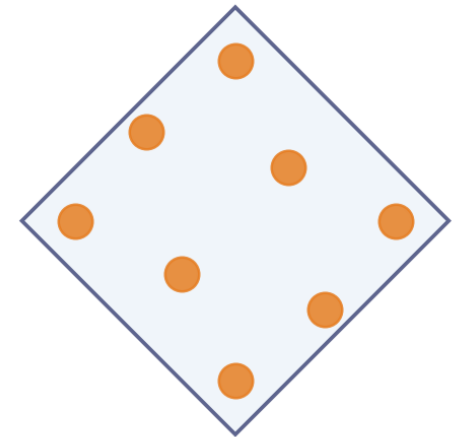
CCD, Box-Behnken

Mixture



simplex, extreme vertices

Optimal



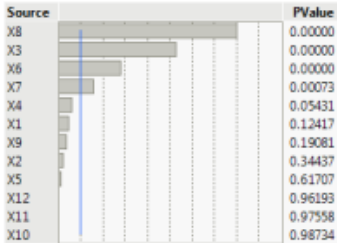
D-optimal, I-optimal

Different designs answer different questions. The tutorial later will let you feel the difference between three of them.

The DOE extended family

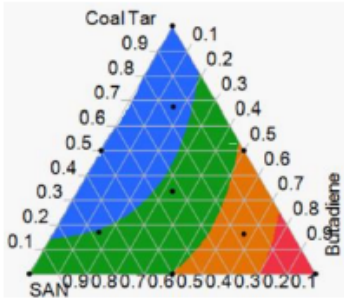
Screening

How to efficiently reduce a large number of possible factors to a few most critical to the process.



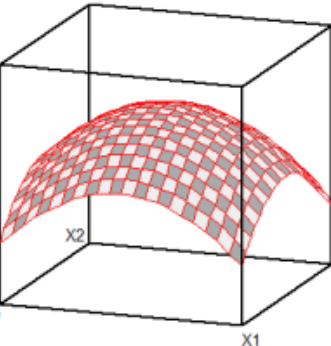
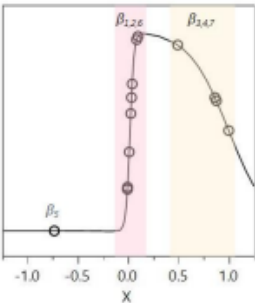
Mixture

For samples that are formulations, how to properly incorporate the dependency in factors due to sum constraint.



Nonlinear

How to select the best design points to estimate a model whose effects are not linear in the parameters.



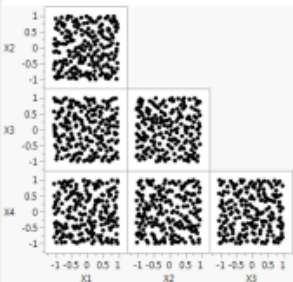
Response Surface

For a few continuous factors, how to incorporate higher order effects like curvature to optimize the response.

	X1 (Hard)	X2 (Easy)	X3 (Easy)
Day 1	Low Level	Low	High
	High Level	High	Low
Day 2	Low Level	High	High
	High Level	Low	Low
Day 3	Low Level	Low	High
	High Level	High	Low

Split Plot

How to alter the sampling plan to account for hard to change factors while controlling the impact of secondary factors.

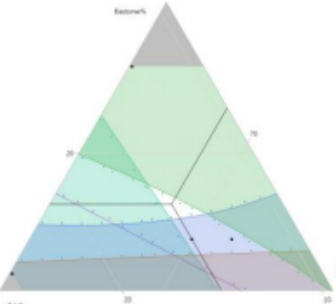


Space Filling

In the presence of minimal measurement error, how to properly ensure coverage of the input space to accurately capture the model form.

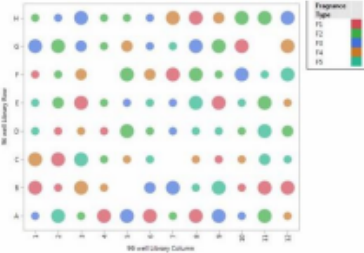
Optimal

How to incorporate factors of varying types (continuous, categorical, mixture) or constraints on factor levels efficiently



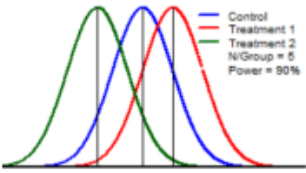
High Throughput

How to properly randomize high throughput experiments to remove row, column, plate effects



Comparison

How to determine the appropriate number of replicates at each treatment level to properly compare response distributions



Choosing factor ranges

THE NUMBERS YOU PICK SCOPE WHAT YOU CAN LEARN

- **Wide enough that the effect is bigger than the noise**
A 2 °C span probably won't show a temperature effect. Use your domain knowledge combined with your goal.
- **Narrow enough to stay physically sensible**
25 → 200 °C might cross a phase change, a decomposition, or a solubility limit you didn't plan for.
- **Bounded by what's actually operable**
Reactor limits, safety, handling.
- **Bounded by chemistry reality**
Consider range-finding experiments to identify boundaries of infeasible regions – design your corners just inside these boundaries. If you skip this step, run your DOE, and 50% of your runs don't produce results, you will waste more time than the range-finding takes.

Constraints and forbidden regions

- **Some combinations are mathematically impossible**

e.g. 100% solvent A AND 100% solvent B cannot coexist

- **Some are operationally forbidden**

e.g. high temperature + high pressure past the reactor rating

- **Some are chemically nonsensical**

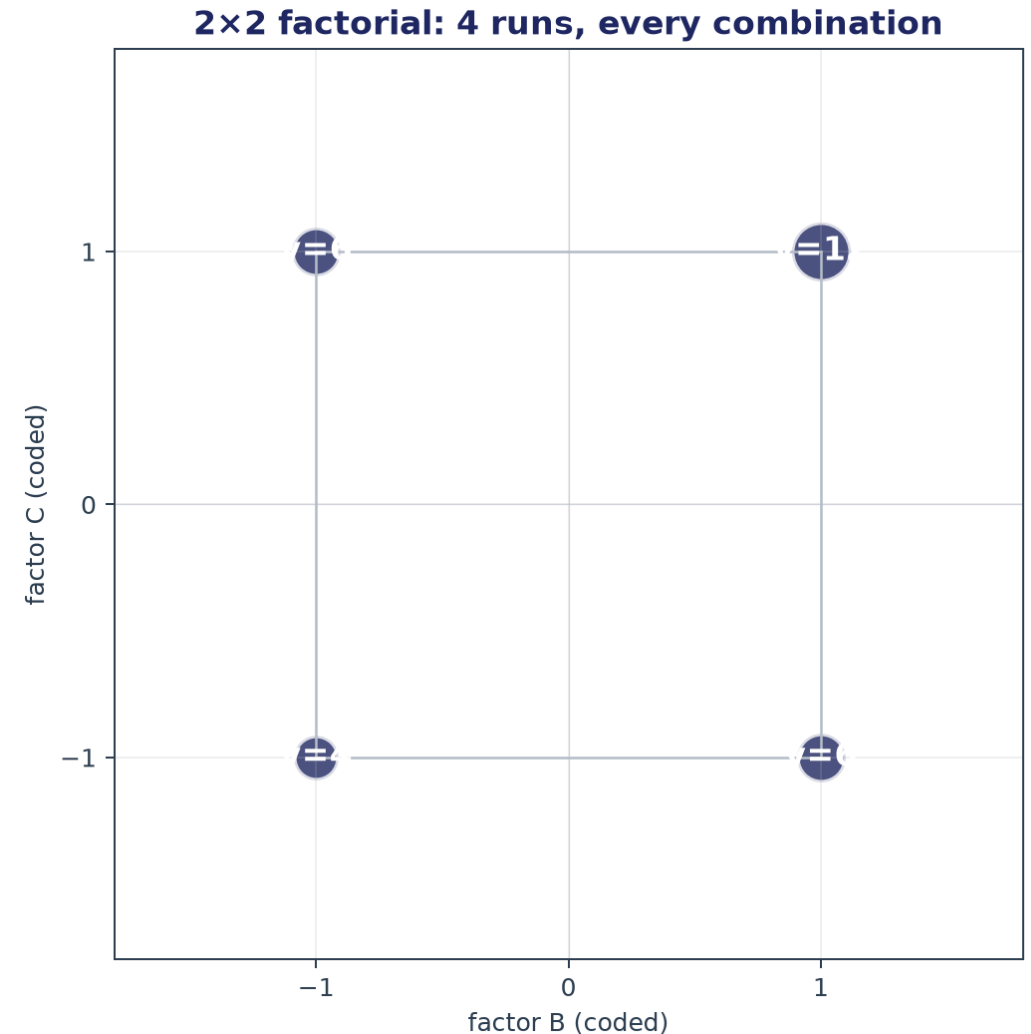
e.g. crosslinker = 0 with cure temp > 150 °C — nothing cures

➤ **What do we do with data points that produced no results because they were not testable?**

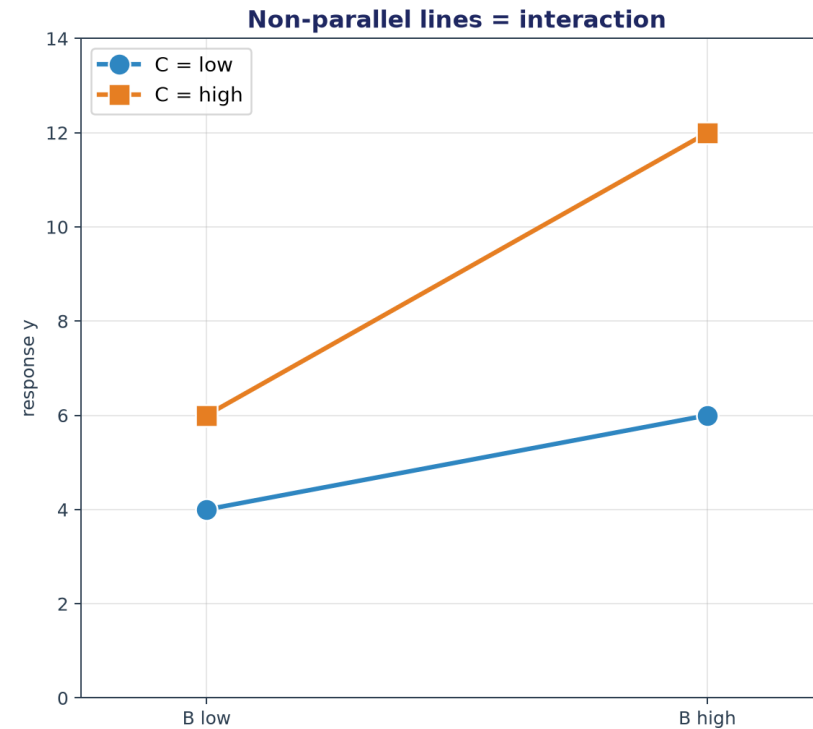
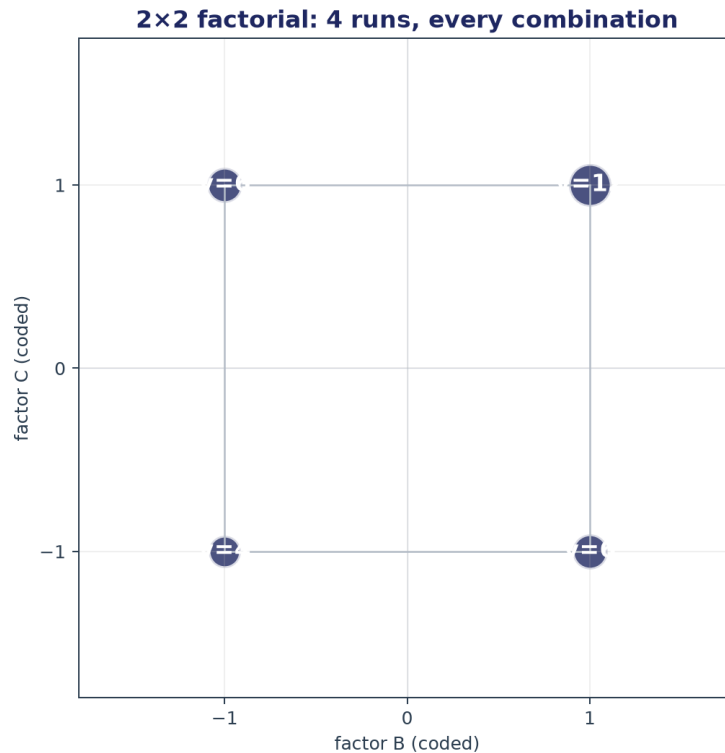
2^K full factorial – screening what matters

A full factorial design runs every combination of factor levels; when used for screening (2 levels) it tests every corner of your k-dimensional hypercube.

Screening seeks to answer the question: “which factors can I ignore because they *don't* matter?” This design cleanly estimates every main effect and interaction with no aliasing, but the run count explodes as 2^k , so once you're past ~ 5 factors it becomes impractical and fractional factorials or other screening designs (Plackett-Burman, definitive screening) are better.



Thinking about interactions

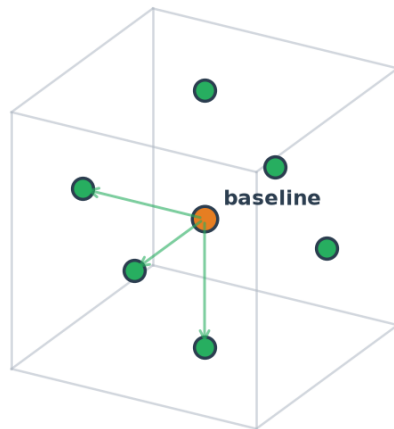


The results of the four corners of a 2×2 factorial, plotted as two lines — one line per level of one factor. If the lines are parallel, the factors don't interact. If they aren't, the gap between their slopes IS the interaction.

Tutorial preview: the three shapes teams will run

OFAT

one factor at a time, from a baseline



orange = baseline

green = single-factor sweep

Fractional factorial

corners of the 2^k design cube

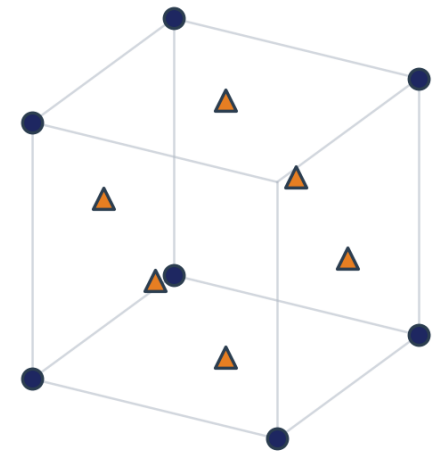


navy = 2^4 corner (± 1 on each factor)

some teams also add center replicates for LOF

Central Composite Design (CCD)

corners + axials — fits a full quadratic



navy = corner (from $2^{(k-1)}$ half-fraction)

orange $\triangle$ = axial 'star' point ($\alpha=1$, face-centered)

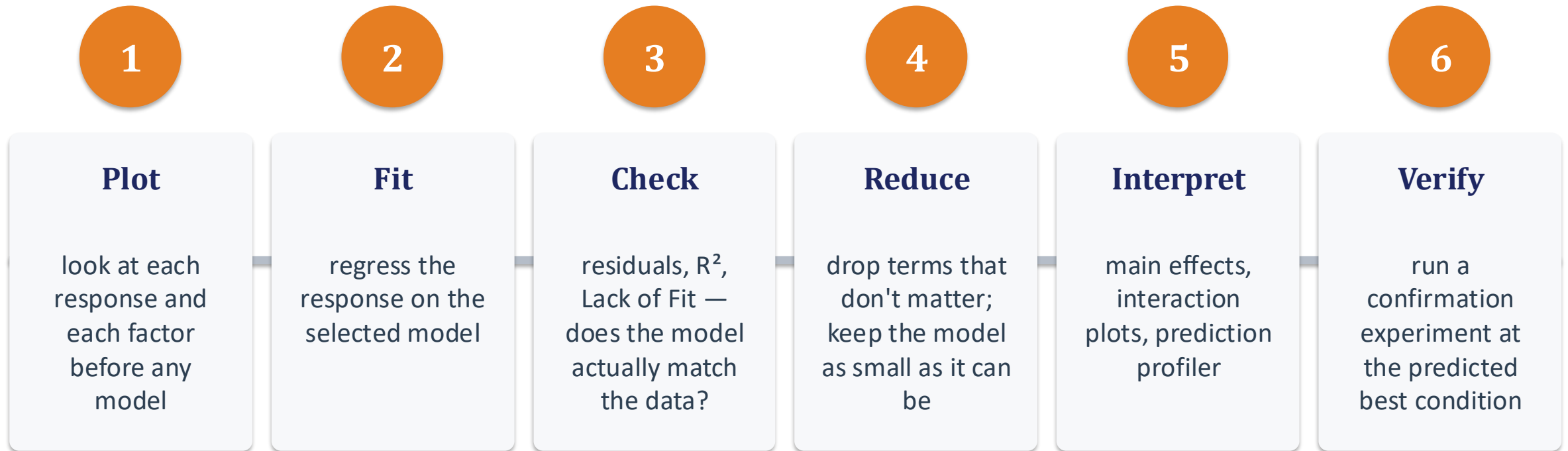
Two new bits of vocabulary: a CORNER is a run where every varied factor is at ± 1 ; an AXIAL point is a run where one factor is at ± 1 and the others are at their midpoints. Corners let you estimate interactions; axials let you estimate pure-quadratic curvature.

PART 3

Analyzing your results

From a table of runs to a defensible answer

A DOE analysis workflow



Is the difference real?

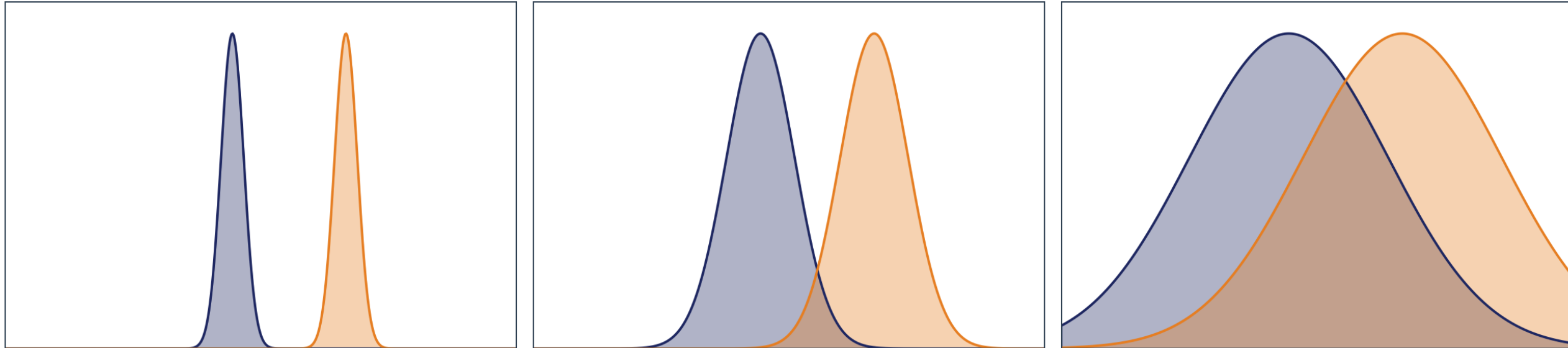
Given that every measurement has noise, is the difference between conditions bigger than what noise alone would produce? That's what comparing groups statistically is designed to answer — with a t-test, ANOVA, or an F-test on regression coefficients.

$$H_0: \beta_1 = \beta_2 = \cdots \beta_i = 0$$

FOR EXAMPLE

Two catalysts, three runs each. Catalyst A yields 68, 71, 69 percent. Catalyst B yields 72, 74, 71 percent. Different means, but the spreads overlap. How confident can we be that the difference isn't noise?

Comparing groups — signal vs within-group spread



The test that answers 'is the difference real' compares two variances: the variance BETWEEN group means and the variance WITHIN each group. If the between-group signal is much bigger than the within-group noise, the difference is real. If not, the noise could explain the pattern. This is the F-ratio at the heart of ANOVA. Every regression coefficient's p-value is a variation on the same idea: this coefficient's size vs the residual noise.

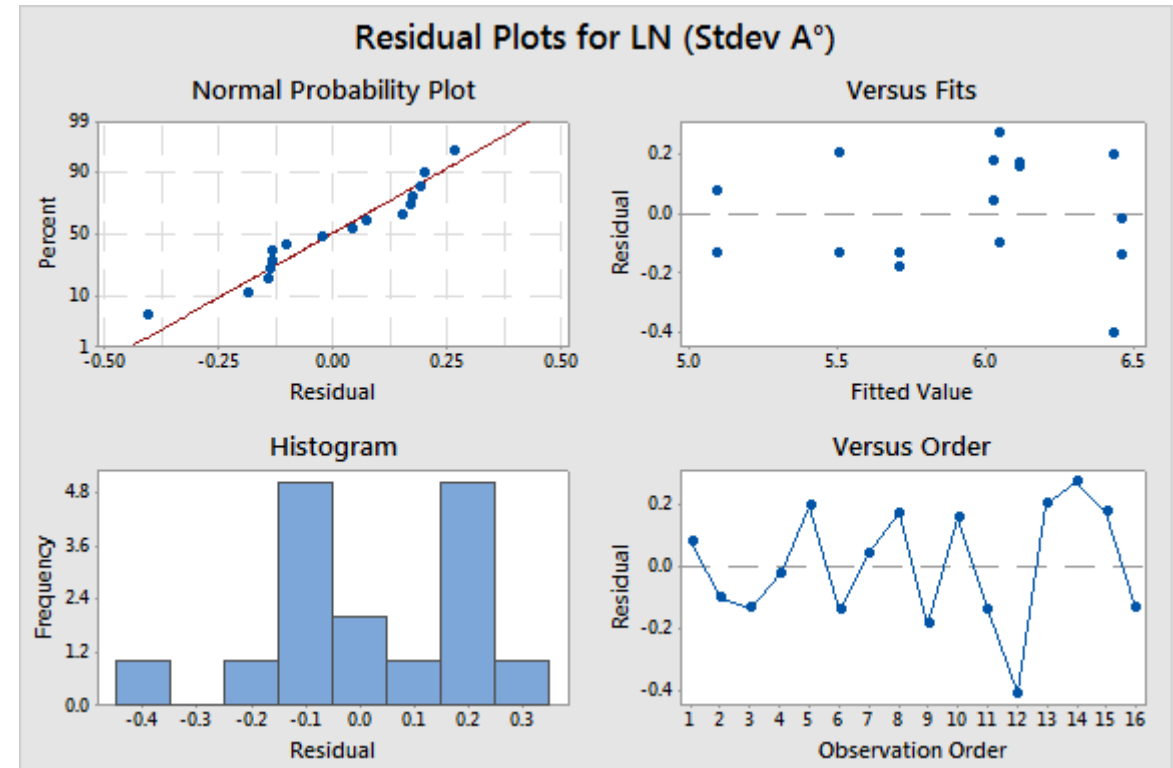
Analysis of DOE results uses the probabilistic linear regression toolkit you learned previously

Step 1: Analysis of Variance (ANOVA) is the workhorse

Additional Steps:

- Visualize residuals and test for normality (you shouldn't see any patterns)
- Lack-of-fit test – are you missing curvature?
- Trim insignificant variables (with hierarchy for interaction terms), re-fit model
 - Overall model F-test
- Identify and check high-leverage points

[NIST Engineering Statistics Handbook - 5.4: Analysis of DOE data](#)



[A DOE in a Manufacturing Environment \(Part 2\)](#)

Statistical significance and practical significance are different questions

Statistical significance

- **Question**
could noise alone have produced this?
- **Answer**
a p-value, a confidence interval, a test statistic
- **Depends on**
the size of the effect AND the number of runs
- **Failure mode**
a tiny effect with a huge sample looks significant

Practical significance

- **Question**
is the effect big enough to change my decision?
- **Answer**
a judgment about what matters for the process
- **Depends on**
the cost of the change and the value of the improvement
- **Failure mode**
a huge naïve effect with too few runs or too large may not be statistically significant

Before you claim you're done — the verification experiment

1. Pick your predicted-best condition

the point on the fitted surface where your objective is optimized

2. Run 2-3 confirmation experiments at that condition

not one — you need to see the noise around the point too

3. Compute the mean and a confidence interval

against the model's prediction and its own confidence interval

4. If they agree — you can trust the research

the fitted model predicts observed reality at the predicted best

5. If they don't — something is off

extrapolation, missed factor, changed process, drift since the DOE. Investigate before you scale up.

AN ASIDE

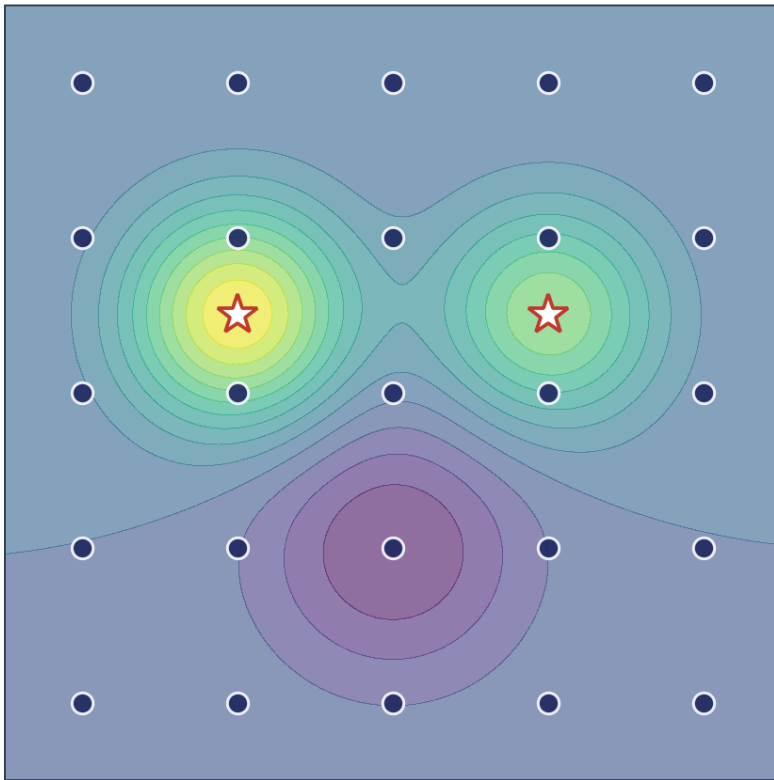
Comparing Experimental Approaches

DOE vs. Active Learning

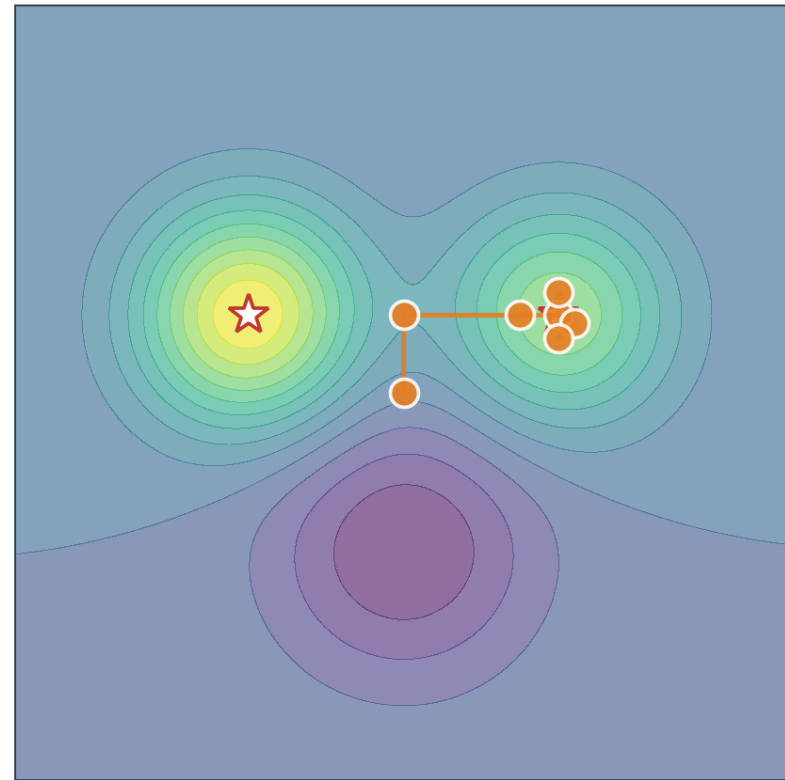
DOE vs. Adaptive Sampling

Same landscape, different sampling strategies — different things you can say about it

DOE grid — samples the whole space



Adaptive sampling — zooms in on one peak



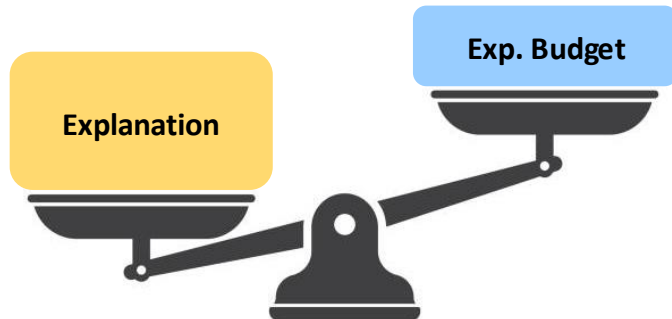
DoE vs. BO: “No free lunch”

Similarity: utilize simulation or experiment results, help to guide the next step experiment(s)

Differences:

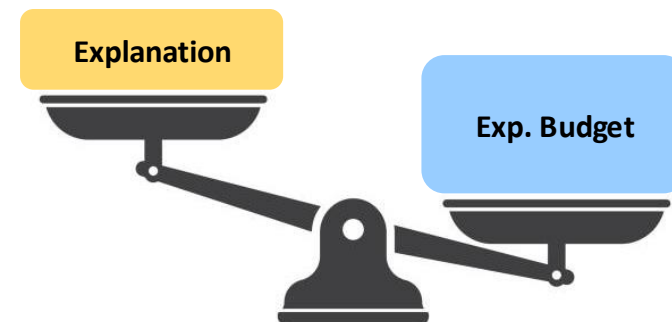
DOE: *Designed Experimentation*

- Build quality model to help screen or optimize
- Explore the entire input space to build a good model, applicable for future studies
- More explanation possible, but require more experimental budget
- Provide a predictive model in the design space



BO: *Bayesian Optimization*

- You want to optimize your reaction
- Provide adaptive and autonomous balance between exploration and exploitation
- Find best solutions faster, but sacrifice interpretation
- Limited predictive power in a small subset of the design space



DOE and active learning: different tools for different questions

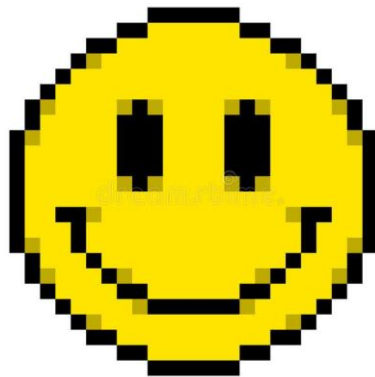
	DOE asks	Active learning asks
Objective	“Tell me what drives what”	“Point me at the next experiment”
Coverage	Maps the whole response surface	Concentrates near the current best guess
Explanation	Explainable structure by definition	Predictive point, not necessarily interpretable
Uses prior data?	No – designs initialize dataset	Yes – Bayesian model uses priors and updates each round
Best for	Understanding, hypothesis testing	Hunting a single optimum on a tight budget
Downside	Can waste runs in uninteresting areas	No information on causality, can miss local optima

Objective: Where is the left eye in the picture?

Experimental budget: Pixels



Actual System



DoE



DFO/BO

When to reach for which

✓ Reach for DOE

- + You need to understand a system**
why does it work; which factors matter
- + You'll explain results to stakeholders**
grows structural knowledge
- + You'll reuse the data later**
map now, exploit later; the map compounds
- + Factors have physical meaning**
worth mapping the whole space you can defend

✗ Reach for active learning

- You have a clear numeric objective**
maximize yield; minimize cost
- You already have some data**
AL uses it as a prior; DOE ignores it
- Budget is tight**
you need the optimum, not the surface
- You don't need to explain WHY**
just WHERE — where's the best next run

Correlation is not causation — and DOE is what turns one into the other

You can find a strong correlation in almost any observational dataset. Sales of ice cream correlate with drowning deaths. Neither causes the other; hot weather causes both.

A designed experiment breaks the ambiguity. When you deliberately vary a factor and measure the response, and everything else is randomized or blocked, a correlation you observe is causal by construction — because you were the one who varied the factor.

This is the deep reason DOE matters: it's the foundation of the scientific method

What to remember from today

Design first

The design step decides what you can learn. If you can't answer 'what model, what runs, what factors?' don't run yet.

Measure your ruler

MSA is not optional. If your ruler is worse than the effect, no design saves you. Bias is not fixable by replication.

Use structure

Using the right design for your question saves time over exploring the space ad-hoc and realizing you missed key relationships later.

Close the loop

Plot, fit, check, reduce, interpret, verify. The analysis is where a designed experiment becomes an answer you can defend.

TAKE A MOMENT

**What would you change about the next
experiment you plan?**

Coming up next — the formulation game

1

Setup

5 min · assign teams,
hand out templates

2

Design + run

30 min · plan and run
15 experiments —
maximize tensile

3

Analyze

15 min · fit your
model, predict your
best

4

Debrief

15 min · reveal plots
show who found the
peak, and why

References and further reading

- [Statistical Thinking for Industrial Problem Solving, presented by JMP | Coursera](#)
- [NIST Engineering Statistics Handbook: Section 5 - Process Improvement](#)
- **Montgomery, D. C.**
Design and Analysis of Experiments (Wiley) — the canonical textbook
- **Box, Hunter & Hunter**
Statistics for Experimenters (Wiley) — chemistry-friendly classic
- **Doyle group**
"Bayesian reaction optimization as a tool for chemical synthesis." Shields et al., Nature 590, 89-96 (2021)
- **pyDOE2, dexpy**
Python packages for design generation
- **statsmodels, scikit-learn**
Python analysis — model fits, ANOVA, regression